Cheon's DLP with auxiliary inputs as an ecbench method

Date: 2026-10-22. Protocol: PROTOCOL.md, frozen before data. Full generated tables: results/results.md (data results/results.json). Follows the P-224-shape study ../cheon_p224_shape_20261022/ (PR #2422).

Summary

- **Methods.** Cheon's n−1 and n+1 algorithms are now registered ecbench methods: `cheon.pminus1_comb`, `cheon.pminus1_naive` and `cheon.pplus1_comb`. They are charged in ecbench's counted unit (`ecbench.gae`, one affine addition or doubling = 1, every phase charged), the same unit as `rho.negation`.
- **The oracle is a separate resource.** The leaked powers [x^i]G come from a planted static-DH oracle (`src/cryptanalysis/ecbench/oracle.rs`). Its queries and the server work behind them are counted outside `total_gae`:
  - `counters.oracle_queries`: d−1 queries for n−1, 2d−1 for n+1;
  - `oracle_server_group_ops`.

  A method sees only `query` and the count. It never sees the scalar, and a public target is refused.
- **Bounds and frontier.** A bound for a querying arm must be fitted with `bound fit --oracle LABEL`, which puts it in an *oracle domain*. There it never sits beside a plain-ECDLP bound, and the frontier reads an extra `oracle_queries` axis. All 46 earlier bound records re-derive unchanged.
- **Data.** 320 runs over 7 curve sizes (32–56 bits) in 3 sessions at L0, counts only. Every answer was verified, and `ecbench verify --replay-all` reproduced 320 of 320 runs exactly.
- **Distance to rho.** Pooled over 36–56 bits, the comb arms cost 0.552 (n−1) and 0.279 (n+1) of matched rho in charged operations at w = 0. The order holds at every price in the 12-cell band (AGENTS.md §5a, extended to oracle queries).
- **No exponent separation is established.** The fitted Cheon exponents (0.450, 0.428) and rho's (0.496, 0.513) have overlapping 95 % intervals.
- **Scope.** This is a cost under a stated oracle (DLP with auxiliary inputs). It is not a plain-ECDLP advance and not an index-calculus improvement.

Sealed bounds

`gae = C·r^α`, where r is the group order. "ops × floor" is the mean S over sqrt(π/2A). The intervals are 95 % bootstrap. The tier is medium (36–56 bits); the 32-bit toy size is reported but never pooled.

curves	arm	α	ops × floor	sizes, runs
n−1	rho.negation	0.496 [0.431, 0.611]	1.114 [0.846, 1.408]	6, 48
n−1	cheon.pminus1_comb	0.450 [0.406, 0.473]	0.648 [0.514, 0.818]	6, 48
n−1	cheon.pminus1_naive	0.441 [0.388, 0.497]	13.814 [10.62, 16.93]	5, 40
n+1	rho.negation	0.513 [0.386, 0.636]	1.143 [0.840, 1.457]	6, 48
n+1	cheon.pplus1_comb	0.428 [0.389, 0.460]	0.657 [0.483, 0.845]	6, 48

- **Naive session reference.** Its own rho reference has α = 0.524 [0.417, 0.685] and ops × floor 1.190.

- **Unpriced native work** at w = 1 GAE per operation:

	n−1	n+1
comb bound	1.074	0.795
rho	2.227	2.285

  At w = 0.1 the comb bounds are 0.690 and 0.670, against rho's 1.225 and 1.257.

- **Oracle fits.** Queries = C·r^α with α = 0.154 [0.092, 0.213] for n−1 and 0.163 [0.146, 0.189] for n+1. The server-operation exponents are 0.189 [0.126, 0.239] and 0.198 [0.178, 0.228].

Gap to matched rho (distance remaining)

Each value is a pooled paired ratio of sums over 36–56 bits, with a 95 % bootstrap over pairs. w is the price of each unpriced native operation, the same on both arms. The oracle price q ∈ {0, 0.1, 1, server work} moves every pooled ratio by at most 0.003, so it is not shown as a column.

arm	w = 0	w = 0.1	w = 1
comb n−1 (48 pairs)	0.552 [0.455, 0.651]	0.528 [0.436, 0.622]	0.418 [0.347, 0.492]
comb n+1 (48 pairs)	0.279 [0.216, 0.419]	0.259 [0.200, 0.388]	0.165 [0.129, 0.247]
naive n−1 (40 pairs)	7.660 [6.33, 9.93]	6.983 [5.77, 9.05]	3.938 [3.26, 5.10]

Per size, comb arms. "Resolved" means the ratio is on the same side of 1 in all 12 price cells.

bits	n−1	n+1
32	dearer than rho	dearer
36	cheaper, interval excludes 1	unresolved
40	dearer	cheaper
44	cheaper, interval excludes 1	cheaper
48	cheaper, interval excludes 1	cheaper
52	cheaper, interval excludes 1	cheaper, interval excludes 1
56	cheaper, interval excludes 1	cheaper, interval excludes 1

- At 56 bits the 12-cell range is 0.484–0.645 for n−1 and 0.142–0.241 for n+1.
- The naive arm is 3.4–27 times dearer than rho at every size and every price.

Frontier. In every oracle domain, rho and the Cheon comb are both on the frontier: Cheon wins on operations and rho wins on oracle queries. The naive arm is dominated by rho.

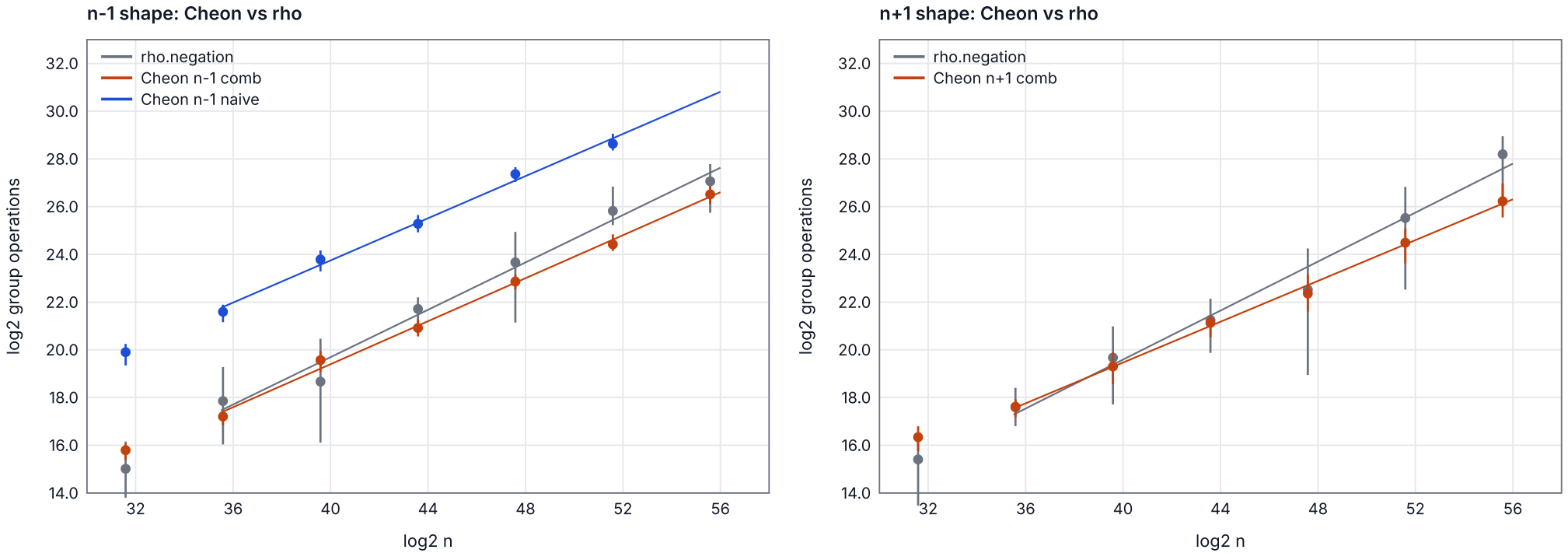


Fig. 1. MEASURED log2 charged operations per curve (dots = mean over verified targets, bars = min..max), ecbench sessions of research/cheon_ecbench_20261022. Lines: the sealed bound fits gae = C r^alpha over 36 to 56 bits (the 32-bit toy size is outside the fit). Oracle queries are not in these operations.

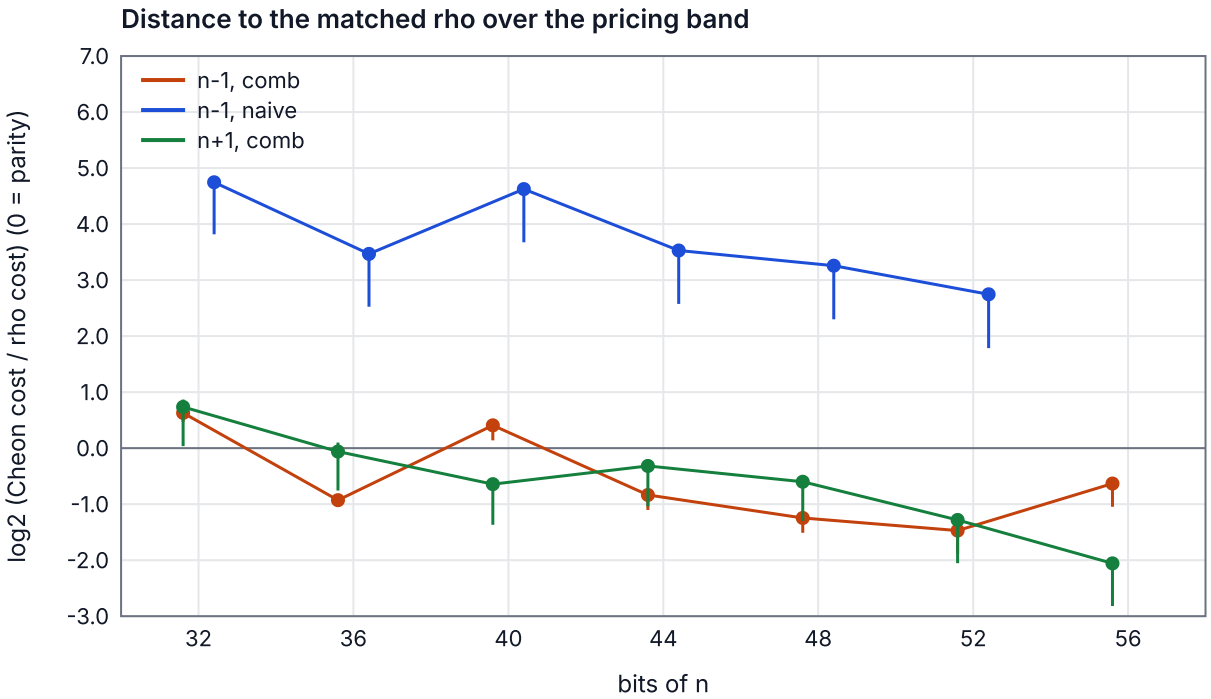


Fig. 2. log2 of the Cheon/rho cost ratio per size. Dots: charged operations only (w = 0, q = 0). Bars: the full range over the 12 pricing cells (w in {0, 0.1, 1} per unpriced native operation, q in {0, 0.1, 1, server work} per oracle query, the same w for both arms). Above 0 = Cheon dearer. Where a bar crosses 0 the order is unresolved. Source: results.json table2.

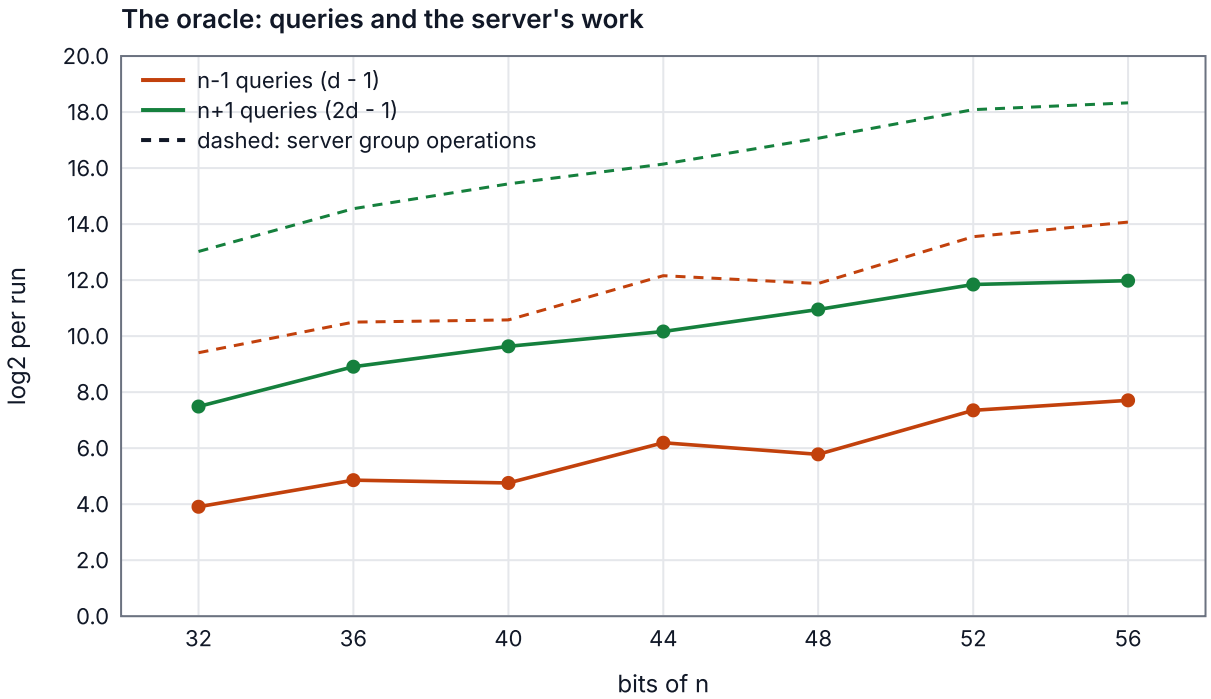


Fig. 3. MEASURED mean static-DH oracle queries per run (solid) and the group operations the server spends answering them (dashed), by size. These are outside the charged operations of Fig. 1. Source: counters.oracle_queries and oracle_server_group_ops of the Cheon comb arms.

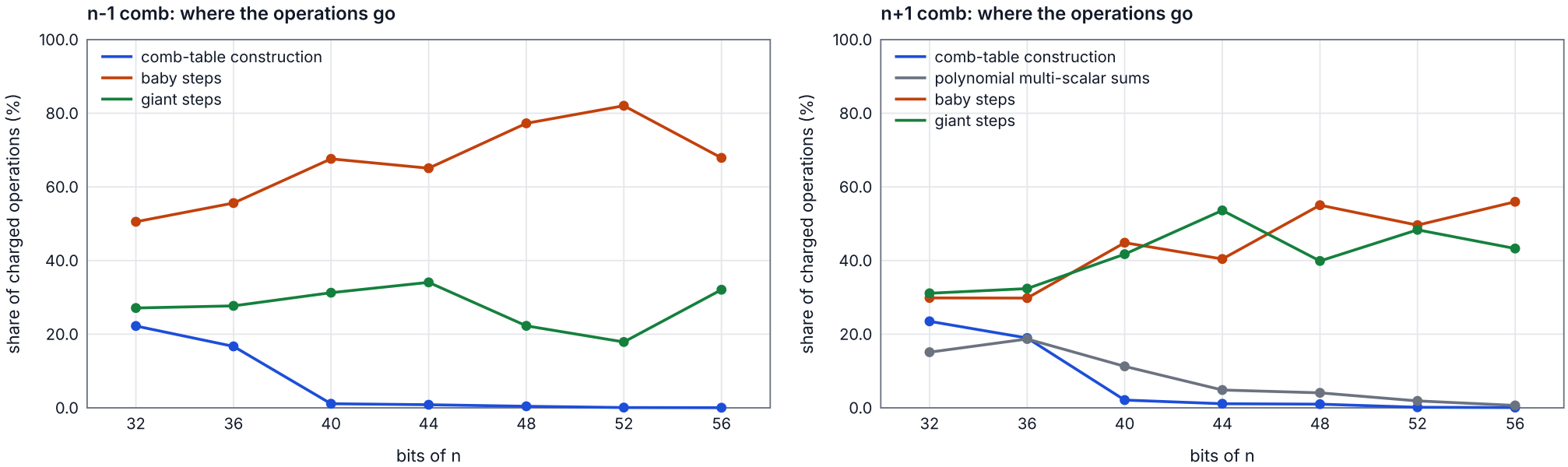


Fig. 4. MEASURED mean share of the charged operations by phase group, Cheon comb arms (phase names containing the group word; the shift and verification phases are the remainder). Source: the phases of each record.

Oracle cost

- **Queries per run:** 29–209 for n−1 and 479–4,031 for n+1.
- **Server work relative to the attacker's work:**

	36 bits	56 bits
n−1	0.95 %	0.018 %
n+1	11.7 %	0.39 %

- **At the P-224 shape** (formula, not measured), the leak needs:

	queries	server operations	attacker cost
n−1	2^29	≈ 2^37	2^97.5
n+1	2^48	≈ 2^56	2^88.4

Pre-registered predictions (PROTOCOL §5)

- **P1, met:** 320 of 320 replays are identical.
- **P2, met:** 120 runs are identical to PR #2422's driver records and none differ. The 6 unmatched runs are the 56-bit naive arm, by design.
- **P3, met:** the fitted exponents (0.450, 0.428; rho 0.496, 0.513) are inside their predicted ranges.
- **P4, partly missed:** the n−1 query exponent 0.154 is inside [0.10, 0.17]. The n+1 query exponent 0.163 is outside [0.17, 0.26], although its interval overlaps that range.

- **P5, met** at 56 bits.
- **P6, met.**
- **Rho sanity gate, met:** S is between 0.60 and 1.50, and the exponents are 0.50–0.52.

Deviations

There are no deviations in curves, d, seeds, targets, unit or pricing band. Two things were added after the freeze, and neither chooses among results:

- the pooled-gap table (Table 2b in `results/results.md`);
- a fix to a figure-legend placement.

The naive arm stops at 52 bits. A single, uncommitted timing probe on the 56-bit curve measured about 8 minutes per target, and PROTOCOL.md discloses this.

The 14 shaped curves have a source record in `curves.json`, but they are **not** yet in `docs/curves/registry.json`. Registering them grows the catalog to 2,586 models. The cover-transfer assessment builder and verifier hard-assert 2,572, so refreshing them is held for a separate decision.

Limits

- **Provenance.** Sessions record `git_commit: null`, because the binary was copied out of the build tree. The binary's sha256 is `0750e00f…f6e5`, built from the `2881fd08e` sources.
- **Host.** Four cores shared with other jobs, at L0. No wall time is reported, and there is no independent-runner receipt.
- **Precision.** With 8 targets per size the intervals are wide, so exponent claims would need more targets.
- **No challenge.** No challenge or verdict was sealed in an oracle domain.

Rho gate: per-run reading (disclosure, 2026-10-23)

PROTOCOL.md section 6 states this study's rho gate per size: *"if the rho `S` is outside `[0.3, 4]` on any size … nothing is ranked against rho"*. It was frozen before any data. Every per-size rho `S` lies between 0.60 and 1.50, so the gate as written is met, and the rankings above follow the protocol.

The protocol also calls the gate "PR #2422's". The review of PR #2422 held that gate to apply to every individual run. Read that way, it fails here too:

session	rho runs	outside `[0.3, 4]`	lowest S
minus	56	7	0.078
minus-naive	48	6	0.078
plus	56	6	0.035

The minus-naive session shares six of its rho targets with the minus session. Rho's per-run cost has a heavy lower tail, as in PR #2422. The pooled ratios above are ratios of sums over 8 paired targets per size. They are therefore less sensitive to single low rho runs than per-size medians, but they are not immune. Whether these rankings are admitted under the per-run reading is a review decision. This section records the counts so that the decision can be made.